Exercise Sheet 6

Evaluation and Empirical Risk Minimisation

Probabilistic Machine Learning

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6.A Measuring a Rare Class

Exercise 6.A.1: A New Fraud Model

A new fraud model is tested on **200,000** transactions, one in 1,000 of them fraudulent. It flags **750** transactions, **150** of which are frauds.

1. Fill in the confusion matrix (TP, FN, FP, TN).
2. Compute the model's accuracy, recall and precision, and the accuracy of the "always legitimate" baseline.
3. A check costs €5, a missed fraud €500. Compute the total cost of the model, of "always legitimate" and of "flag everything".
4. The model is moved to a market where one transaction in **100** is fraudulent. Its recall and its false-alarm rate stay the same. What is its precision there?

Exercise 6.A.2: The Bayes Error

A single attribute x takes three values:

x	$P(x)$	$P(y = 1 \mid x)$
A	0.5	0.1
B	0.3	0.5
C	0.2	0.8

1. Which classifier has the smallest true error (0/1 loss)? Compute its error — the Bayes error.
2. What is the error of "always predict 0"?
3. A team reports a test error of **0.20** on **100** test cases. Is that evidence that their model beats the Bayes error? What if the same 0.20 came from **10,000** test cases?
4. Would more rows of the same data lower the Bayes error? What would?

6.B Numbers With Error Bars

Exercise 6.B.1: Error Bars on a Recall

On a test set of **50,000** transactions there are **50** frauds; model A catches **40** of them.

1. Compute the recall, its standard error and a ±2 SE interval.
2. With a flat prior, what is the Beta posterior for the recall? Give its mean. (Its 95% credible interval is $[0.67, 0.89]$.)
3. How many test frauds — and roughly how many transactions — are needed for an error bar of ±0.02 (±2 SE) at a recall near 0.8?
4. Model B catches 42 of the 50. The two models disagree on **5** frauds: B caught 4 of them, A caught 1. If both models were equally good, each disagreement would go either way with probability ½. How likely is a split at least as lopsided as 4 : 1 in B's favour? Conclusion?

6.C From Scores to Decisions

Exercise 6.C.1: Thresholds From Costs

A check costs €5 (c_{FP}), a missed fraud €500 (c_{FN}).

1. Derive the threshold p^* above which checking is cheaper in expectation.
2. The bank buys chargeback insurance, and a missed fraud now costs **€100**. What is the new threshold?
3. Four transactions have calibrated scores 0.002, 0.03, 0.09 and 0.4. Which are checked under the old and under the new costs? Verify the 0.03 case with expected costs.
4. It turns out the model's scores are **twice** the true probabilities. What goes wrong with the rule of part 1, and how do you repair it?

Exercise 6.C.2: Scoring Probabilities

Five held-out transactions with outcomes $y = (1, 0, 0, 0, 1)$ (1 = fraud).

	p_1	p_2	p_3	p_4	p_5
model A	0.8	0.1	0.2	0.1	0.6
model B	0.99	0.01	0.01	0.01	0.4

1. Compute each model's accuracy at the threshold 0.5.
2. Compute each model's log-loss (natural logarithm) and Brier score.
3. The three measures do not agree. What does each one reward?

6.D The Shortcut: Keep the Loss

Exercise 6.D.1: What Is ERM, and What Does It Drop?

1. Show that the MAP estimate for linear regression with Gaussian noise $\mathcal{N}(0, \sigma^2)$ and prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \tau^2\mathbf{I})$ minimises $\frac{1}{n}\sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 + \lambda \|\mathbf{w}\|^2$. Express λ in σ, τ and n.
2. Evaluate λ for $\sigma = 40, \tau = 10$ and $n = 20$, then $n = 2{,}000$. What happens to the prior?
3. For each method, say whether it is ERM in the form $\min_h \frac{1}{n}\sum_i L(h(\mathbf{x}_i), y_i)$, and whether its output is a probability: logistic regression · linear SVM · naive Bayes (maximum likelihood) · Bayesian linear regression.